UNIT-2

①

Gradually Varied Flow

Dynamic equations of gradually varied and spatially varied flow- Water surface flow, profile classifications: Hydraulic Slope, Hydraulic curve- Profile determinations by Numerical method, Direct step method and standard step method, Graphical method - Applications.

Dynamic Equations :-

Gradually varied flow occurs in many situations like upstream of a barrier, downstream of a sluice gauge, due to change of slope of a channels.

To study this flow and its profiles, knowledge of normal depth and critical depth is essential.

$$\frac{dh}{dx} = \frac{i_b - i_e}{1 - [V^2/gh]}$$

$$= \frac{i_b - i_e}{1 - [F_e]^2}$$

i_b → Slope of the channel head

i_e → Slope of energy line

F_e → Froude Number.

As h is the depth of flow and x is the distance measured along the bottom of the channel. Hence $\frac{dh}{dx}$ represents the variation of water depth along the bottom of the channel.

When $\frac{dh}{dx}$ is equal to zero, h is constant (or) depth of water above the bottom of the channel is constant. It means that

free surface of water is ||le to the bed of the channel, when dh/dx is greater than zero (ie) +ve means the depth of water ↑ in the direction of flow. The profile of water so obtained is called backwater curve.

When dh/dx is less than zero, (ie) −ve, means depth of water ↓ in the direction of flow. The profile of water so obtained is called draw down curve.

Dynamic Equations of Spatially Varied flows :-

Spatially Varied flow :-

Flow varies with longitudinal distance is called spatially varied flow.

Momentum and energy equations are used to develop dynamic equations for spatially varied flow. The energy equation is more appropriate for lateral outflow and momentum equation is suitable for lateral inflow.

$$\frac{dy}{dx} = \frac{S - S_e - \left[\frac{\alpha Q q}{g A^2}\right]}{1 - \alpha F^2}$$

Dynamic equation for spatially varied flow with decreasing discharge.

$$\frac{dy}{dx} = \frac{S - S_e - \left[\frac{2\alpha Q q}{g A^2}\right]}{1 - \alpha F^2}$$

Dynamic equation for spatially varied flow with increasing discharge.

Classification of water surface flow profiles :-

Based on slope of channel bed (S), normal depth of water (y_0) and critical depth (y_c), the channels are classified into five categories.

If the flow in the channel is uniform, the channel is said to have a normal slope denoted by S_n.

Mild Slope :- (M)

If the slope of the channel bottom (S) is less than the critical slope S_c ($S < S_c$), the channel slope is called mild slope.

In channels of the mild slope, the normal depth is always greater than the critical depth, (ie) $y_0 > y_c$

Critical Slope :- (C)

If the slope of the channel bottom (S) is equal to the critical slope S_c ($S = S_c$), the channel slope is called critical slope. In such channels, the normal depth is equal to the critical depth, (i.e) ($y_0 = y_c$)

Steep Slope :- (S)

If the slope of the channel bottom (S) is more than the critical slope S_c ($S > S_c$), the channel slope is called steep slope. In such channels, the normal depth is always less than the critical depth, (i.e) [$y_0 < y_c$ (supercritical)].

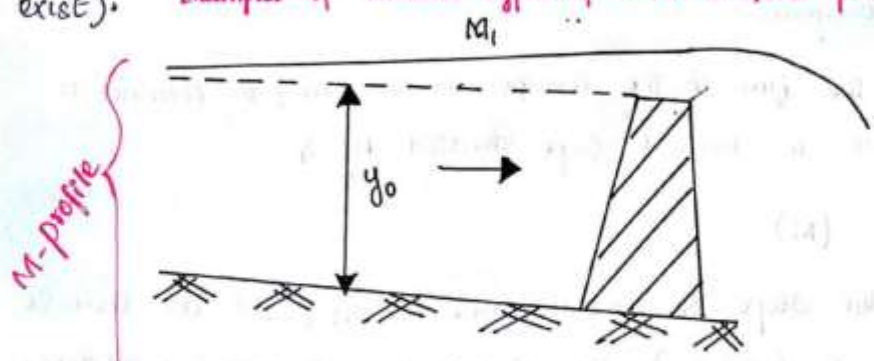
Horizontal bed :- (H)

When the channel bottom is hz. (ie) $S = 0$, is called horizontal bed. The normal depth tends to be infinite for such channels, (i.e) $y_0 = \infty$ (y_0 does not exist).

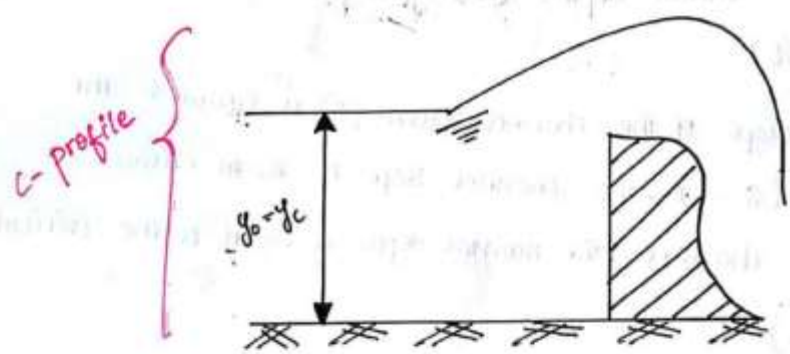
Adverse slope :- (A)

When the bottom channel rises in the direction of flow ($S<0$) is called adverse slope. It is called negative slope (y_0 does not exist).

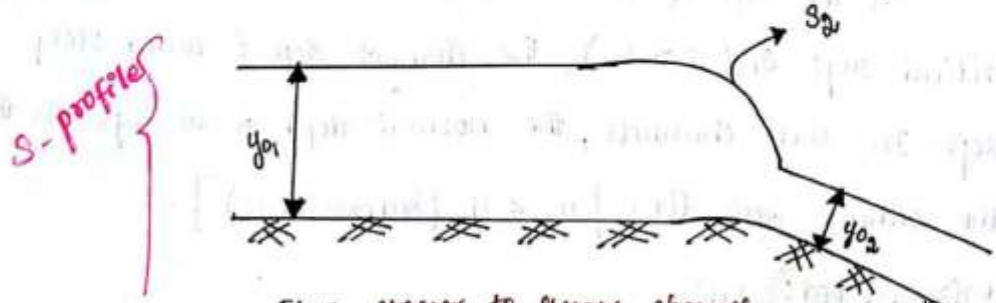
Examples of various types of water surface profiles.



(a) Flow behind an overflow weir.

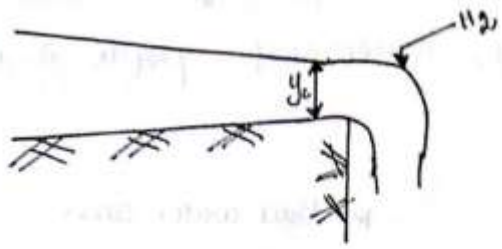


Flow behind an overflow weir



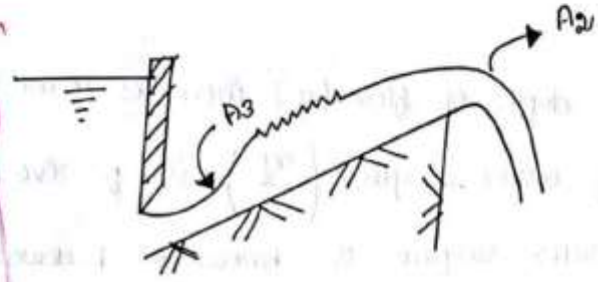
Flow steeper to steeper channel.

H-type profile.



Flow over a free overfall

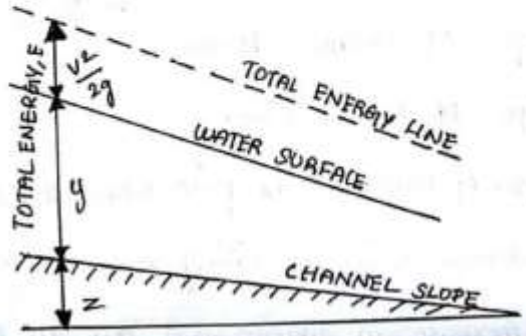
A-type profile



Flow downstream of a sluice with adverse slope and terminating in a free over fall.

Hydraulic Slope :-

Hydraulic slope / Hydraulic Gradient line (HGL) is merged with a free water surface, since there is no hydraulic pressure head in open channel flow.



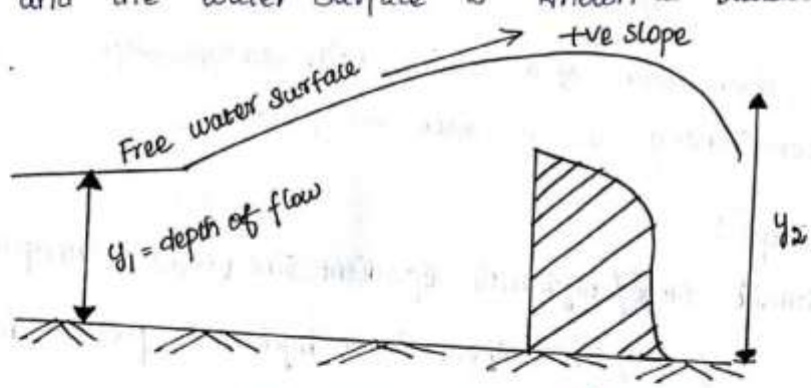
Backwater and Drawdown curves or hydraulic curves.

Based on depth of water, the flow direction and slope of water surface, water surface profile is classified in to two types

Water Surface Profile. → Back water curve
Water Surface Profile. → Draw down curve

Back water Curve :-

When the depth of flow (y) increases in the direction of flow, slope of water surface $\left(\frac{dy}{dx}\right)$ is positive (upward slope) and the water surface is known as backwater curve).



S_w = Slope of water surface $= \left[\frac{dy}{dx}\right]$

S = Slope of channel bottom.

S_e = Slope of energy line.

y_1 = Depth of water at the point where the water starts rising up

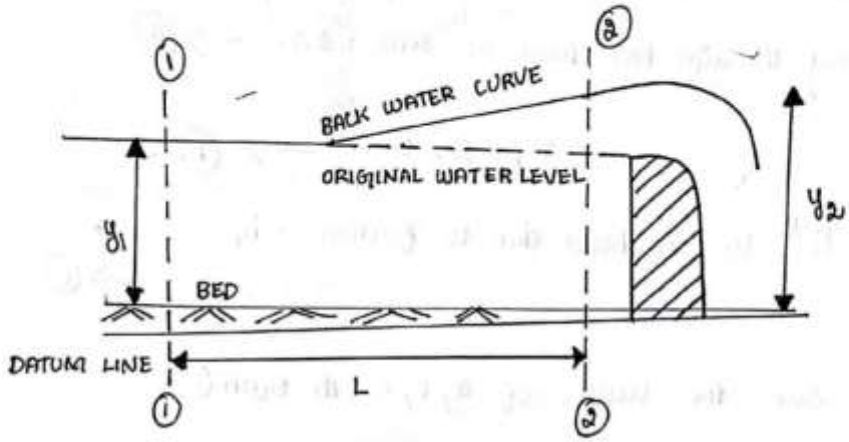
y_2 = Maximum height of rising water from bed.

The maximum increase in water level due to obstruction in the path of flow of water is called afflux $[y_2 - y_1]$.

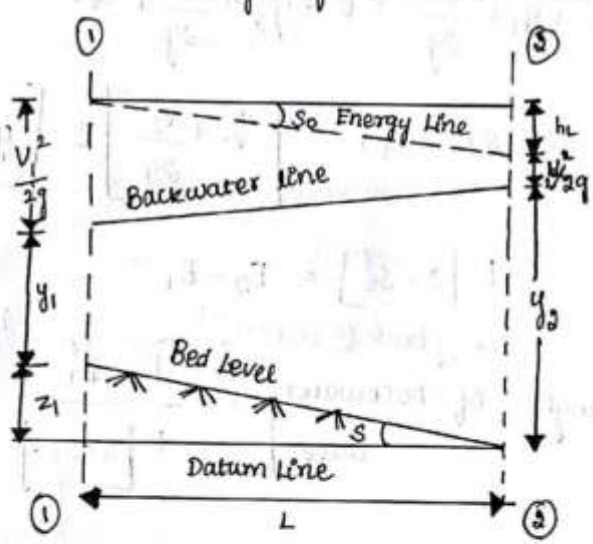
Expression for the length of Backwater curve :-

(4)

Consider the flow over a dam. Due to obstruction (dam), the water level raises and it has a maximum depth of water near the dam.



Length of back water curve.



y_1 = Depth of flow at section 1-1

y_2 = Depth of flow at section 2-2

V_1 = Velocity of flow at section 1-1

V_2 = Velocity of flow at section 2-2

S = Bed slope

S_e = Energy line slope

L = Length of back water curve.

Applying Bernoulli's equation at section 1-1 and section 2-2

$$z_1 + y_1 + \frac{V_1^2}{2g} = z_2 + y_2 + \frac{V_2^2}{2g} + h_L \rightarrow (1)$$

Datum head at section 2-2, $z_2 = 0$, since datum line is passing through bed level at section 2-2. $\rightarrow$ (a)

$$z_1 = S \times L \rightarrow (b)$$

$\|^{ly}$ Loss of head due to friction $= h_L$

$$= S_e \times L \rightarrow (c)$$

Sub the values of a, b, c in eqn (1),

$$SL + y_1 + \frac{V_1^2}{2g} = 0 + y_2 + \frac{V_2^2}{2g} + S_e L$$

$$SL - S_e L = \left[y_2 + \frac{V_2^2}{2g}\right] - \left[y_1 + \frac{V_1^2}{2g}\right]$$

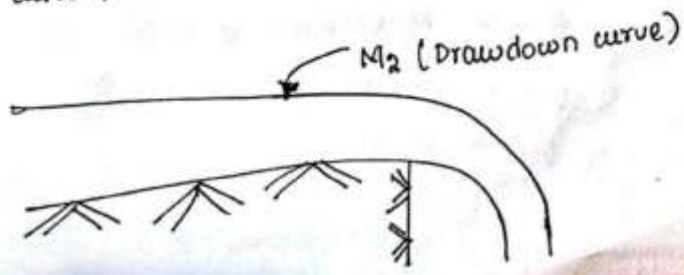
$$L\left[S - S_e\right] = E_2 - E_1$$

Energy head @ section 1-1 Energy head @ section 2-2.

$$\text{Length of backwater curve} = \frac{E_2 - E_1}{L\left[S - S_e\right]}$$

Slope of energy line.

Draw down curve :-



(5)

When the depth of flow decreases along the flow direction, $\frac{dy}{dx}$ becomes negative and the surface profile is called draw down curve.

Profile determination by numerical method :-

Length of surface profile is determined with the help of any one of the following methods.

(a) Graphical Integration method.

(b) Direct Step method.

(c) Standard step method.

Graphical Integration method :-

The differential equation of the gradually varied flow may be inverted and integrated to give the length of the surface profile b/n depths y_1 and y_2. Hence the equation for gradually varied flow is gn by,

$$\frac{dy}{dx} = \frac{S - S_e}{1 - F^2}$$

$$\frac{dx}{dy} = \frac{1 - F^2}{S - S_e}$$

$$dx = \left[\frac{1 - F^2}{S - S_e}\right] dy$$

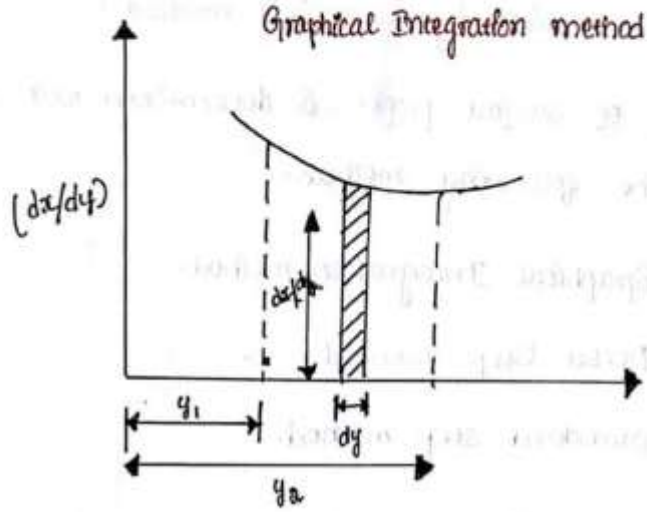
Integrating on both sides,

$$\int dx = \int \left[\frac{1 - F^2}{S - S_e}\right] . dy$$

It may be written as,

$$x = L = \int_{y_1}^{y_2} \left[\frac{1-F^2}{S-S_0}\right] \cdot dy \longrightarrow (2)$$

Graphical Integration method.



By plotting $\frac{dx}{dy}$ against y, and computing the area under the curve enclosed by the y-axis and the ordinates of $\frac{dx}{dy}$ corresponding to depths y_1 and y_2.

Direct Method :-

The channel is divided into short distances and computations are carried out b/n two ends. Two ends are mentioned as sections (1-1) and (2-2). Bernoulli's equation is used to calculate the length of backwater. It is called as step by step method. It is applicable to prismatic channels only.

Standard Step method :-

Standard step method is used for non-prismatic/natural channels. In natural channels, the cross-section varies

from section to section and also the cross-section infm. (b) is known only at a few locations along the channel.

The c/s information at two adjacent sections and the discharge and datum at one section are required to determine the datum at the other section.

The solution of the above problem is obtained by a trial and error method from the basic energy equation.

Applications :-

Direct step method, standard step method, and graphical methods are used for the determination of length of the surface profile for gradually varied flows (GVF).

For GVF problems, the channel is divided into short distances and computations are carried out from one end of the reach to the other.

In direct step method, Manning's formula is sufficient to accurately evaluate the slope of total energy line.

The standard step method is mainly applicable to non-prismatic channels, ex: natural river. A trial and error method is employed in this method.

Problems :-

① Find the slope of the free water surface of a rectangular stream 10m wide and 2m deep. The slope of the stream bed is 1 in 5000. Total discharge is $20m^3/s$. Assume C = 65 and that the depth is increasing in the direction of flow.

Given Data :-

Width of channel, b = 10m.

Depth of water, y = 2m.

Slope of bed, $S = \frac{1}{5000}$

$= 0.0002$.

Discharge (Q) = $20m^3/s$.

Chezy's constant, C = 65.

Solution :-

S_e = Slope of energy line can be calculated with the help of Chezy's formula.

$$Q = AV$$
$$= by \times C\sqrt{RS_e}$$

wetted perimeter (P) = $b + 2y$
$= 10 + (2 \times 2)$
$= 14m$.

wetted area (A) = $b \times y$
$= 10 \times 2$
$= 20m^2$.

(7)

Hydraulic mean radius,

$$R = \frac{\text{Wetted Area}}{\text{Wetted Perimeter}}$$

$$= \frac{20}{14}$$

$$= 1.43\,m.$$

$$20 = 10 \times 2 \times 65 \times \sqrt{1.43 \times S_e}$$

$$S_e = 1.655 \times 10^{-4}$$

$$= \frac{1}{6041.75}$$

Velocity of flow,

$$V = \frac{Q}{A} = \frac{20}{10 \times 2} = 1\,m/s.$$

Slope of water surface,

$$\frac{dy}{dx} = \frac{S - S_e}{1 - \frac{V^2}{gy}}$$

$$= \frac{(0.0002 - 0.0001655)}{1 - \left[\frac{1^2}{9.81 \times 2}\right]}$$

$$= 3.635 \times 10^{-5}$$

$$\left.\begin{array}{l}\text{Slope of free water}\\ \text{surface}\end{array}\right\} = \frac{1}{27508}$$

(2) A river 100m wide and 3m deep has stable bed and vertical banks with a surface slope of 1 in 3000. Estimate the length of backwater curve produced by an afflux of 2m. Assume Manning's $n = 0.035$.

Given data :-

Width of river, $b = 100m$.

Depth of water @ section 1-1, $(y_1) = 3m$.

River surface slope of bed, $(S) = \frac{1}{3000}$

$= 0.333 \times 10^{-3}$

Manning's $(n) = 0.035$

Chezy's constant, $(C) = 65$

Afflux, $(y_2 - y_1) = 2m$.

Solution :-

Wetted perimeter at section 1-1,

$$P = b + 2y_1$$
$$= 100 + (2 \times 3)$$
$$= 106m.$$

Wetted area @ section 1-1,

$$A = b \times y_1$$
$$= 100 \times 3$$
$$= 300 m^2.$$

Hydraulic mean radius,

$$R = \frac{\text{Wetted area}}{\text{Wetted Perimeter}} = \frac{300}{106} = 2.83m.$$

Manning's formula, (8)

$$Q = A\frac{1}{n}(R)^{2/3}(S)^{1/2}$$

$$= 300 \times \frac{1}{0.035} \times (2.83)^{2/3}\left(\frac{1}{3000}\right)^{\frac{1}{2}}$$

$$= 313.11\ m^3/s$$

Specific energy @ 3m, depth,

$$E_1 = y_1 + \frac{V_1^2}{2g}$$

$$V_1 = \frac{Q}{A} = \frac{313.33}{300}$$

$$= 1.044\ m/s.$$

$$E_1 = 3 + \frac{(1.044)^2}{2 \times 9.81}$$

$$E_1 = 3.056\ m.$$

Depth of section @ section [2-2], $y_2 = 2 + y_1$

$$= 2 + 3$$

$$= 5m.$$

Specific energy @ 5m depth,

$$E_2 = y_2 + \frac{V_2^2}{2g}.$$

$$A_2 = by^2 = 100 \times 5 = 500\ m^2$$

$$V_2 = \frac{Q}{A_2} = \frac{313.33}{500}$$

$$= 0.626\ m/s.$$

$$E_2 = 5 + \frac{(0.626)^2}{2 \times 9.81} = 5.02m.$$

Energy line slope at the mean section where the depth,

$$y = \frac{3+5}{2} = 4m.$$

Wetted Perimeter,

$$P = b + 2y = 100 + (2 \times 4) = 108m.$$

Wetted Area,

$$A = b \times y = 100 \times 4 = 400m^2.$$

Hydraulic mean radius,

$$R = \frac{\text{Wetted Area}}{\text{Wetted Perimeter}} = \frac{400}{108}$$

$$= 3.70m.$$

From Manning's formula,

$$Q = AV = A \cdot \frac{1}{n} (R)^{2/3} (S_e)^{\frac{1}{2}}$$

$$313.11 = 400 \times \frac{1}{0.035} \times (3.7)^{2/3} (S_e)^{1/2}$$

$$S_e = 1.311 \times 10^{-4}$$

$$= \frac{1}{7625}.$$

Length of backwater curve,

$$\Delta x = \frac{E_2 - E_1}{S_b - S_e} \quad [\because S_b = \text{Bed slope}]$$

$$\Delta x = \frac{5.02 - 3.056}{\frac{1}{3000} - \frac{1}{7625}}$$

$$= 9713.8m.$$

The normal depth of flow of water, in a rectangular channel 2m wide is 1.2m. The bed slope of the channel is 0.0006 and Manning's roughness co-efficient $n = 0.015$. Find the critical depth. (a)

At a certain section of the same channel, the depth is 0.9m while at a second section the depth is 0.85m. Find the distance b/n two sections (Use one reach in the calculations). Also find whether the second section is located downstream/upstream with respect to the first section.

Given data :-

Width of channel, $b = 2m$.

Depth of water, $y = 1.2m$.

Slope of bed, $S = 0.0006$

$= \frac{1}{1666.67}$

Manning's co-eff (n) $= 0.015$

At section 1-1, depth of water $y_1 = 0.90m$.

At section 2-2, depth of water $y_2 = 0.85m$.

Solution :-

(a) Critical depth calculations,

Wetted perimeter $(P) = b + 2y$

$= 2 + (2 \times 1.2)$

$= 4.4m$.

Wetted area, (A) = b×y

$= 2 \times 1.2$

$= 2.4 m^2$

Hydraulic mean radius,

$$R = \frac{\text{Wetted area}}{\text{Wetted perimeter}} = \frac{2.4}{4.4}$$

$= 0.55 m.$

Manning's formula,

$$Q = A \cdot \frac{1}{n} (R)^{\frac{2}{3}} (S)^{1/2}$$

$$= 2.4 \times \frac{1}{0.015} \times (0.55)^{2/3} (0.0006)^{1/2}$$

$= 2.631 m^3/s$

$$q = \frac{Q}{b} = \frac{2.631}{2}$$

$= 1.316 m^3/s$

Critical depth @ section 2-2,

$$y_c = \left[\frac{q_2^2}{g}\right]^{1/3}$$

$$= \left[\frac{(1.316)^2}{9.81}\right]^{1/3}$$

$= 0.561 m.$

(b) Distance b/n two sections calculation.

Specific energy at 0.90m depth section (1-1)

(10)

$$E_1 = y_1 + \frac{V_1^2}{2g}$$

$$V_1 = \frac{Q}{b_1 y_1} = \frac{2.631}{2 \times 0.9}$$

$$= 1.461 \, m/s$$

$$E_1 = 0.90 + \frac{(1.461)^2}{2 \times 9.81}$$

$$= 1.0009 \, m.$$

Specific energy at 0.85m depth (section 2-2).

$$E_2 = y_2 + \frac{V_2^2}{2g}$$

$$V_2 = \frac{Q}{b_2 y_2} = \frac{2.631}{2 \times 0.85}$$

$$= 1.548 \, m/s$$

$$E_2 = 0.85 + \frac{(1.548)^2}{2 \times 9.81}$$

$$= 0.972 \, m.$$

Using Manning's formula for energy line slope calculation

Mean depth of water,

$$y = \frac{1}{2}(y_1 + y_2)$$

$$= \frac{1}{2}(0.90 + 0.85)$$

$$= 0.875 m.$$

Wetted perimeter,

$$P = b + 2y = 2 + (2 \times 0.875) = 3.75 m.$$

Wetted Area, (A) = bxy

$= 2 \times 0.875$

$= 1.75 m^2$.

$$\text{Hydraulic mean radius} = \frac{\text{Wetted Area}}{\text{Wetted Perimeter}}$$

$$= \frac{1.75}{3.75} = 0.467 m.$$

Manning's formula,

$$Q = A \cdot \frac{1}{n} \cdot (R)^{2/3} (S)^{1/2}$$

$$2.63 = 1.75 \times \frac{1}{0.015} \times (0.467)^{2/3} (Se)^{1/2}$$

$$Se = 1.404 \times 10^{-3}$$

$$= \frac{1}{712.43}$$

Distance b/n two sections,

$$\Delta x = \frac{E_2 - E_1}{S - Se}$$

$$= \frac{0.972 - 1.009}{(6 \times 10^{-4}) - (1.404 \times 10^{-3})}$$

$$= 46.02 m.$$

If free water surface slope $\left[\frac{dy}{dx}\right]$ is negative, the second section is downstream and $\left(\frac{dy}{dx}\right)$ is +ve, the second section is upstream.

(11)

Water surface slope,

$$\frac{dy}{dx} = \frac{S - S_e}{1 - \frac{V^2}{gy}}$$

Average depth of flow $(y) = 0.875m$.

$$V = \frac{Q}{A} = \frac{2.631}{2 \times 0.875}$$

$$= 1.503 \, m/s.$$

$$\frac{V^2}{gy} = \frac{(1.503)^2}{9.81 \times 0.875}$$

$$= 0.263$$

$$\frac{dy}{dx} = \frac{(6 \times 10^{-4}) - (1.404 \times 10^{-3})}{1 - 0.263}$$

$$= -1.09 \times 10^{-3}$$

$$= -\frac{1}{916.67}$$

Since $\frac{dy}{dx}$ is (-ve), the second section is downstream

(4) A rectangular flume 1.5m wide carries discharge at the rate of $2m^3/s$. The bed slope of the flume is 0.0004. At a certain section, the depth of flow is 1m. Calculate the distance of the section downstream where the depth of flow is 0.75m. Solve by single slope method. Assume rugosity co-efficient as 0.014.

Is the slope of the channel mild / steep? How is this type of surface profile classified?

Given data :-

Discharge, $Q = 2m^3/s$.

Width of channel, $(b) = 1.50m$.

Bed slope, $(S) = 0.0004$

$= 0.4 \times 10^{-3}$

At section 1-1, depth of flow $= 1m$.

At section 2-2, depth of flow $= 0.75m$.

Rugosity co-efficient = Manning's co-efficient

$n = 0.014$.

Solution :-

(a) Discharge per unit width, $(q) = \frac{Q}{b}$

$= \frac{2}{1.5}$

$= 1.33 m^2/s$

critical depth, $(y_c) = \left[\frac{q^2}{g}\right]^{1/3} = \left[\frac{1.33}{9.81}\right]^{1/3}$

critical depth of flow, $(y_c) = 0.565m$.

Hydraulic mean depth, $(R_c) = \frac{A_c}{P_c}$

$= \frac{(1.5 \times 0.565)}{(1.5 + 2 \times 0.565)}$

$= 0.322m$.

Using Manning's formula, the critical slope,

$$Q = A_c \cdot \frac{1}{n} (R)^{2/3} \cdot (S_c)^{1/2}$$

$$2 = [1.5 \times 0.565] \times \left[\left[\frac{1}{0.014}\right] \times (0.32)^{2/3}\right] \times (S_c)^{1/2}$$

$$\boxed{S_c = 4.94 \times 10^{-3}}$$

$$\text{Critical slope} = 0.00495 = \frac{1}{202.02}$$

Since $S < S_o$, the channel is mild, and the water surface is falling, it is a M-2 type curve.

(b) Energy line slope at the mean section where the water depth,

$$y = \frac{y_1 + y_2}{2}$$

$$y = \frac{1 + 0.75}{2} = 0.875 m.$$

$$R = \frac{A}{P} = \frac{(1.5 \times 0.875)}{(1.5 + 2 \times 0.875)}$$

$$= 0.404 m.$$

For energy line slope (S_e), using Manning's formula

$$Q = A_c \frac{1}{n} (R)^{2/3} (S_e)^{1/2}.$$

$$2 = (1.5 \times 0.875) \left(\frac{1}{0.014}\right) (0.404)^{2/3} (S_e)^{1/2}$$

$= 1.524 \times 10^{-3}$.

Velocity, $(V_1) = \frac{Q}{A_1} = \frac{2}{(1.5 \times 1)} = 1.33$ m/s.

Ily

$$V_2 = \frac{Q}{A_2} = \frac{2}{(1.5 \times 0.75)} = 1.78 \text{ m/s.}$$

Specific energy @ section 1-1,

$$E_1 = y_1 + \frac{V_1^2}{2g}$$

$$E_1 = 1 + \frac{(1.33)^2}{2 \times 9.81}$$

$$= 1.091 \text{ m.}$$

Specific energy @ section 2-2,

$$E_2 = y_2 + \frac{V_2^2}{2g}$$

$$= 0.75 + \frac{1.78^2}{2 \times 9.81}$$

$$= 0.91 \text{ m.}$$

As per direct step method,

$$\Delta x = \frac{E_2 - E_1}{S - S_e}$$

$$= \frac{0.91 - 1.091}{(0.4 \times 10^{-3}) - (1.524 \times 10^{-3})}$$

Distance b/n section 1-1 & 2-2 $= 161.03$ m.

(13) Determine the slope of the free water surface in a rectangular channel of width 20m, having depth of flow 5m. The discharge through the channel is 52 cumecs. The bed slope of the channel is 1 in 4000. Assume chezy's constant C as 60.

Given Data :-

Width of channel, $b = 20m$.

Depth of flow, $y = 5m$.

Discharge, $Q = 52\ m^3/s$.

Slope of channel bed $(S) = 1$ in 4000

$= \frac{1}{4000} = 2.5 \times 10^{-4}$.

chezy's constant $= 60$.

Solution :-

Hydraulic radius $(R) = \frac{A}{P} \Rightarrow m = \frac{b \times y}{b+2y}$

$= \frac{20 \times 5}{20+(2 \times 5)} = \frac{100}{30}$

$= 3.33m$.

Discharge, $Q = A \times V$

$= AC\sqrt{RS}$

$52 = 100 \times 60\sqrt{3.33 \times S_e}$

$S_e = 2.25 \times 10^{-5}$

Velocity, $V = \frac{Q}{A}$.

$= \frac{52}{20 \times 5} = 0.52\ m/s$.

The slope of free water surface.

$$\frac{dy}{dx} = \frac{S - S_e}{1 - F^2}$$

For rectangular section, hydraulic mean depth

$$D = \frac{A}{T} = y.$$

$$\frac{dy}{dx} = \frac{S - S_e}{1 - \left[\frac{V^2}{gy}\right]}$$

$$= \frac{(2.5\times10^{-4}) - (2.25\times10^{-5})}{1 - \left[\frac{(0.52)^2}{9.81\times5}\right]}$$

$$= 2.287\times10^{-4}$$

$$= \frac{1}{4372}$$

A trapezoidal channel with bed width of 10m and side slopes 1 : 1.5 is carrying a flow of 80 m^3/s. The channel bottom slope is 0.002 and Manning's 'n' is 0.015. A dam is planned in such a way that the flow depth increases to 10m. Determine the depth of flow in the channel 250m, 500m and 750m upstream of the bed.

Given data :-

Bottom width of trapezoidal channel, b = 10m.

Depth of flow at (dam) point 'A', y_A = 10m.

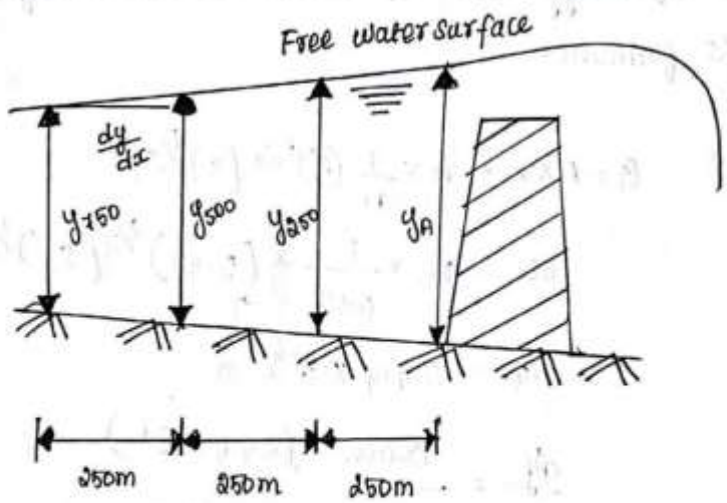
Side slope of channel = 1.5H : 1V, m = 1.5

Discharge Q = 80 m^3/s.

Manning's constant, n = 0.015

Bed slope of the channel = 0.002.

(14)



Longitudinal section of a trapezoidal channel.

$$R = \frac{\text{Wetted Area}}{\text{Wetted Perimeter}} = \frac{A}{P}$$

$$\text{Area } (A) = (b + my_A)\, y_A$$

$$= (10 + 1.5 \times 10)\, 10$$

$$= 250\, m^2.$$

$$\text{Wetted Perimeter, } P = b + 2y_A\sqrt{m^2+1}$$

$$= 10 + 2 \times 10\sqrt{1.5^2+1}$$

$$= 46.05\, m.$$

$$R = \frac{250}{46.05}$$

$$= 5.43\, m.$$

Slope of free water surface,

$$\frac{dy}{dx} = \frac{S - S_e}{1 - \left[\frac{V^2}{gD}\right]}$$

Slope of energy line, S_e can be calculated by using Manning's formula.

$$Q = A \times V = A \times \frac{1}{n} (R)^{2/3} (S)^{1/2}.$$

$$80 = 250 \times \frac{1}{0.015} \times (5.43)^{2/3} (S_e)^{1/2}$$

$$S_e = 2.414 \times 10^{-6}.$$

$$\frac{dy}{dx} = \frac{0.002 - (2.414 \times 10^{-6})}{1 - \left[\frac{(0.32)^2}{9.81 \times 10}\right]}$$

$$= \frac{1.997 \times 10^{-3}}{0.9967}$$

$$= 2 \times 10^{-3}.$$

Depth of flow in the channel 250m from the dam (point 'A')

$$y_{250\,m} = y_A - \left[L \cdot \frac{dy}{dx}\right]$$

$$= 10 - \left[250 \times 2 \times 10^{-3}\right]$$

$$= 9.50\,m.$$

Depth of flow in the channel 500m from the dam (point 'A')

$$y_{500\,m} = y_A - \left[L \cdot \frac{dy}{dx}\right]$$

$$= 10 - \left[500 \times 2 \times 10^{-3}\right]$$

$$= 9\,m.$$

Depth of flow in the channel 750m from the dam (ii)

(Point A').

$$y_{750m} = y_A - \left[L \cdot \frac{dy}{dx}\right]$$

$$= 10 - \left[750 \times 2 \times 10^{-3}\right]$$

$$= 8.50m.$$